A Neuro-Symbolic Framework for Common-Sense Reasoning in Domestic Robots

Evaluation on the RoboCSK Benchmark & Out-of-Distribution Generalization Study

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02/06/2026

RoboCSK - Results Summary

RoboCSK — Summary Dashboard

Task	Format	Metric	Best Score	Best Configuration
Tidy Up	Multi-choice	Accuracy	**0.6997**	mistral-large + Full pipeline
Tidy Up	Open / Generative	MRR	**0.897**	mistral-medium + Pure LLM
Tool Usage	Multi-choice	Accuracy	**0.8179**	mistral-small + Full pipeline
Table Setting	Plate prediction	Accuracy	**0.880**	mistral-small + Full pipeline
Table Setting	Cutlery prediction	Jaccard	**0.778**	mistral-large + Full pipeline
Procedural Knowledge	Binary ordering	Accuracy / F1	**0.983**	mistral-small + Pure LLM
Procedural Knowledge	MC next step	Accuracy	**0.863**	mistral-medium + Pure LLM
Meta-Reasoning	Binary capability	Accuracy / F1	**0.996**	Pure Logic
Meta-Reasoning	MC robot config	Accuracy	**0.943**	Pure Logic

Takeaway: the best setup is task-dependent: Full pipeline leads Tidy Up (MC), Tool Usage, and Table Setting; Pure LLM leads Procedural Knowledge; Pure Logic leads Meta-Reasoning.

Generalization Study

Generalization Datasets

RoboCSK Task	→ OOD Dataset	Domain Shift
Tidy Up	TidyBot, VirtualHome	Real-world robotic cleanup, 3D household simulator
Tool Usage	PIQA	Open-text physical reasoning
Meta-Reas.	BEHAVIOR-1K	1,000 complex activities
Proc. Know.	RecipeNLG	2.2M real human recipes

Tidy Up Generalization

The tidy-up task was evaluated on **two independent OOD datasets** at different levels of realism and scale:

Dataset	Source	Questions	Ground Truth
VirtualHome	3D household simulator	67	Simulator rules
TidyBot	Real robot deployments	598	User preferences

VirtualHome was the **first choice** since it provides a ground truth derived from a simulated domestic environment.

However, **67 questions is too small** for a statistically meaningful ablation study (full 3×3 model $\times$ ablation matrix $= 9$ runs over only 67 samples).

TidyBot was added as a second, larger OOD source, though it introduces a different kind of domain shift: **user-preference-dependent** placement. Meaning that TidyBot locations encode a single user's organisational preferences (e.g. "graphic novels go on the top shelf"). The KB encodes commonsense defaults (e.g. "books go on shelves"). When user preference diverges from commonsense, the pipeline's grounding signal is **misleading by design**.

Generalization: Tidy Up → VirtualHome

(Puig et al., 2018)

What is VirtualHome?

A 3D household simulator where agent programs are executed as sequences of high-level actions. Objects and receptacles have **fixed, deterministic placement rules** encoded in `object_script_placing.json`. **Dataset construction**

Extracted from two simulator files:

- `object_script_placing.json` — which objects are placed in which receptacles
- `properties_data_unity.json` — object properties (grabbable, surface, etc.)

Two filters applied to produce clean MC questions:

1. **Object must be `GRABBABLE`** in Unity — non-portable furniture (bed, sofa, bookshelf) excluded, since a tidy-up robot only relocates portable items
2. **Destination must be a `CONTAINERS` or `SURFACES` receptacle** in Unity — ensures the target is a real storage location, not for example a floor.

Generalization: Tidy Up → VirtualHome

Mapping example

<table>
<tr><th>VH simulator entry</th><th>Generated MC row</th></tr>
<tr><td>

`object_script_placing.json`:

```
"alarm clock": {
  "placing": ["nightstand",
              "kitchencounter",
              "coffeetable"]
}
```

</td><td>

Object: `alarm clock`
Correct: `nightstand`
Wrong (VH vocab):
`garbagecan`
`cuttingboard`
`kitchencounter`
`fridge`

</td></tr>
</table>

Results Analysis

Mode A — Vocab-Aligning Evaluation Names normalised via synonym table + fuzzy matching before KB lookup.

Model	Full Pipeline	Pure LLM	Δ
large	91.04%	**92.54%**	**−1.50**
medium	92.54%	**94.03%**	**−1.49**
small	**88.06%**	88.06%	**0.00**

Mode B — Strict Evaluation No synonym normalisation, location strings must match the KB exactly.

Model	Full Pipeline	Pure Logic	Δ vs. Mode A
large	89.55%	—	**−1.49**
medium	92.54%	—	**0.00**
small	71.64%	34.33%	**−16.42**

Results Analysis

Interpretation

In vocab-aligning mode the pipeline **matches or slightly underperforms** the pure LLM because VirtualHome locations are straightforward commonsense (beer→fridge, alarm clock→nightstand) that a large LLM already knows with high confidence. The symbolic layer adds noise rather than signal when the LLM is already near ceiling (92%).

± 1.5 pp $= 1$ question difference on 67 items. VH confirms the pipeline's core geometry is correct — the challenge is vocabulary bridging.

Tidy Up → TidyBot

(Wu et al., 2023)

Pipeline Pipeline Mapping Example

Raw Source Entry (`scenarios.yml`)	Generated Multi-Choice Row (`tidybot_multichoice.csv`)

<pre>room: bedroom
receptacles:
  - storage chest
  - closet
  - cabinet
  - nightstand
  - top shelf
unseen_placements:
  - - graphic novel 2
    - top shelf</pre>
	Target Evaluation Object:
	`Object: "graphic novel 2"`
	Ground Truth Label:
	`Correct_Location: "top shelf"`
	Seeded Global Vocab Padding ($N = 4$):
	`Wrong_Locations:`
	`['storage chest', 'closet', 'cabinet', 'nightstand']`

Tidy Up → TidyBot

Why is it highly challenging?

- **The Subjectivity Paradox**: Pure common-sense defaults fail (scoring just 45.6% without preferences). System targets are completely dependent on human tastes (e.g., matching a user's explicit `seen_placements`).

Evaluation Results — 598 Multi-Choice Questions

Model	Full Pipeline	Pure LLM Baseline	Δ
large	**47.99%**	39.13%	**+8.86**
medium	41.30%	30.94%	**+10.36**
small	34.45%	26.15%	**+8.30**

ID *to* OOD Domain Shift Performance Drop: −22 pp.
However, the symbolic grounding advantage holds completely robust (+8.3 to +10.4 pp).

Tool Usage Generalisation: Exploring PIQA

The PIQA Benchmark (Bisk et al., 2020)

- Evaluates **physical commonsense reasoning**
- Validation subset used: **1,838 questions**
- Binary choice format using **free-text solution paragraphs**
- Spans everyday activities: cooking, cleaning, DIY, and personal care

Hypothesis & Motivation PIQA shares clear domestic domain commonalities with Robo-CSK (e.g., cutting, wiping). We aimed to test if our KB-affordance grounding pipeline could extend from tool identification to **physically correct procedure selection**.

Domain Optimization An LLM-based classifier isolated **710 household-only questions** to maximize overlap with our knowledge base domain.

PIQA Structural Layout

Goal:
Clean kitchen tiles.

Solution 1 ✅
"Use half slice of potato to clean kitchen tiles, then rinse."

Solution 2 ❌
"Use half slice of peach to clean kitchen tiles, then rinse"

Key Insight: Targets are descriptive procedural paragraphs. Correctness relies on material characteristics and safety, not simple object affordance availability.

Why PIQA Does Not Map to Tool Usage

Expected Setup: Entity Selection

Goal: clean up a spill

Choice A: sponge → `absorb` affordance ✅
Choice B: drill → irrelevant ❌

Result: Pipeline matches affordance and selects **A**.

Structural Alignment Archetype:

- Alternatives are **discrete nouns**
- Correct answer maps directly to a **KB-addressable object**
- The isolated affordance lookup provides an unambiguous signal

Actual Setup: Procedural Reasoning

Goal: make sure a log doesn't roll

- **Sol 1** ✅: "Place a small *wedge* under the log on both sides."
- **Sol 2** ❌: "Place a small *tarp* under the log on both sides."

Goal: remove stripped screws

- **Sol 1** ✅: "Use a *rubber band* between screwdriver and screw for grip."
- **Sol 2** ❌: "Use a *rubber band* around the screwdriver handle."

The Signal Paradox: Physical validity depends on intuitive mechanics and step sequencing. Target words (like `screwdriver`) often show up across **both options**, rendering simple vocabulary presence completely uninformative.

The Geometry Mismatch & Evaluation Outcomes

Structural Comparison Matrix

Dimension	Tool Usage (Robo-CSK)	PIQA
Answer Type	Discrete noun	Free-text procedure
Correct Choice	KB-grounded object	Physically valid method
Distractor	Wrong-affordance item	Implausible method
KB Core Signal	Direct lookup alignment	Noisy substring overlaps

The Geometry Mismatch & Evaluation Outcomes

Household-Filtered Split Results (710 Qs)

Model Tier	Full Pipeline	Pure LLM	Delta
Mistral Medium	91.69%	**94.37%**	**−2.68 pp**
Mistral Large	90.28%	**91.97%**	**−1.69 pp**
Mistral Small	80.56%	**83.66%**	**−3.10 pp**

Ablation Outcome: The symbolic addition systematically degrades performance across all model sizes, highlighting a firm **architectural boundary condition**.

- **Pure Logic Baseline: 49.29%** (equivalent to random guessing), showing keyword matching cannot capture physical plausibility.

Exclusion Analysis & Architectural Boundaries

Rationale for Exclusion

Even when filtered strictly for domestic scenarios, PIQA questions structurally break the core neuro-symbolic framework:

- Core inquiry targets **operational correctness**, not asset selection.
- Symbolic tags often fire as false positives on invalid branches.
- Retaining it within tool evaluation would misrepresent pipeline capabilities.

Diagnostic Value: This exploration is kept as a definitive **boundary-condition test**, mapping the exact perimeter where explicit affordance injection breaks down.

Meta-Reasoning → BEHAVIOR-1K

(Li et al., 2024)

Mapping Example — From Task Text to Binary Evaluation Pair

Source (`behavior1k_tasks.json`) + LLaMA 3.1 Annotation	Generated Binary Row (`behavior1k_annotated.csv`)

`{ "task": "assembling furniture",`
`"category": "maintenance" }`
 	
Step 1 — LLaMA 3.1 annotation :
 	
`{ "mobile": false, "arms": 1,`
`"min_dof": 7, "gripper": "two_finger",`
`"rigid_grip": true }`
 	
Step 2 — Mutation engine: strips `arms → 0`, `DoFs → 0`, `gripper → none`
	Positive (Minimum Viable Robot):
	`Model: "Super_Robot"`
	`Arms: 1, DoFs: 7, Rigid Gripper: ✓`
	`Can execute?: True`
	
	Negative (Counter-factual Mutation):
	`Model: "Super_Robot_No_Arms"`
	`Arms: 0, DoFs: 0, Gripper: none`
	`Can execute?: False`

Meta-Reasoning → BEHAVIOR-1K

Evaluation Results — 2,032 Out-of-Distribution Binary Questions

Model	Full Pipeline	Pure LLM	Pure Logic	Δ Full vs. LLM
large	**0.768 F1**	0.571 F1	0.728 F1	**+0.197**
medium	0.745 F1	0.542 F1	0.728 F1	**+0.203**
small	0.765 F1	0.612 F1	0.728 F1	**+0.153**

Zero-Shot Symbolic Transfer: `reasoner.pl` is applied with **zero modifications** to the meta-reasoning domain. The +19.7 pp F1 gain (large model, Full vs. Pure LLM), with Pure Logic scoring 0.728 F1 independently, confirms the symbolic layer anchors robotic capability reasoning even under harsh OOD task shift.

Procedural Knowledge → RecipeNLG

(Bień et al., 2020)

In-distribution (Robo-CSK)

- Source: **Recipe1M+** corpus
- Pairs generated by **GPT-4o**
- Clean, well-structured language
- Explicit temporal markers

OOD benchmark (RecipeNLG)

- Source: **2.2 M** real home-cook recipes
- Steps written by thousands of different users
- Noisy, informal, fragmented language
- No temporal markers assumed

The Prolog reasoner and LLM prompts are loaded **completely unmodified**, pure zero-shot transfer. No prompt engineering, no fine-tuning.

Generated Dataset size:
500 step pairs → **2,000 binary questions**

RecipeNLG Evaluation Set

Step 1 — Pair extraction

Consecutive recipe steps provide ground-truth ordering:

step $n \rightarrow$ step *(n+1)*
$\Longrightarrow$ step n **always** comes before *(n+1)*

- Max **3 pairs per recipe** (avoid over-representation)

Step 2 — Balanced sampling

From the full corpus:

- Max **2 pairs per unique recipe title**
- **500 pairs** sampled uniformly
- Perfectly balanced: 50 % Yes / 50 % No

Step 3 — 4 questions per pair

Question	GT
Did step 1 occur **before** step 2?	Yes
Did step 2 occur **before** step 1?	No
Did step 2 occur **after** step 1?	Yes
Did step 1 occur **after** step 2?	No

Example: "Easy Sweet n' Sour Pork"

Recipe: Easy Sweet n' Sour Pork
Step A: Blanch the bamboo shoot and slice it into bite-sized pieces.
Step B: Coat the pork in the katakuriko and fry.

Results: Robo-CSK (ID) vs. RecipeNLG (OOD)

OOD — RecipeNLG *(2,000 binary questions)*

Model	Approach	F1
mistral-large	**Full pipeline**	**0.898**
mistral-large	Pure LLM	0.786
mistral-medium	**Full pipeline**	**0.812**
mistral-medium	Pure LLM	0.640
mistral-small	**Full pipeline**	**0.846**
mistral-small	Pure LLM	0.798
-	Pure Logic	0.406

Key Findings: Procedural Knowledge Generalization

Symbolic Layer Amplification Under Domain Shift

Model	Full Pipeline	Pure LLM Baseline	Δ F1
mistral-medium	0.812	0.640	**+0.172**
mistral-large	0.898	0.786	**+0.112**
mistral-small	0.846	0.798	**+0.048**

On the synthetic Robo-CSK benchmark, the neuro-symbolic delta was near-zero (+0.008 for large). On the naturalistic RecipeNLG dataset, **the symbolic lift amplifies significantly up to +17.2 pp**.

Core Analytical Pillars

1. Invariance of Universal Culinary Logic (`reasoner.pl`)

The 11-phase operational hierarchy (preheating → prep → mixing → ... → cooking → ... → serving) encodes hard physical constraints rather than superficial text patterns. Causal physics remain completely invariant across sources: a home cook cannot serve a dish before cooking it, or cook items before preparing them, enabling seamless zero-shot transfer with zero adaptation.

Key Findings: Procedural Knowledge Generalization

2. LLM Recall Collapse vs. Key Word Breakdown

- **The Phrasing Trap**: Pure LLM drop peaks for mistral-medium ($0.927 \rightarrow 0.640$, a collapse of -0.287). Real human steps use highly fragmented context (*"Add water."*, *"Work in flour with hands if sticky."*) that lacks explicit temporal markers, blinding the model.
- **Heuristic Failure**: Pure Logic drops below chance (40.6% vs. 50% baseline) because static keyword matchers overfit to clean verb lists and systematically misclassify colloquial instructions like *"let sit"* or *"do this"*.

3. OOD Generalization Penalty Analysis

The absolute performance drop for mistral-large is a very modest -0.070 (0.968 in-distribution down to 0.898 out-of-distribution). Retaining an F1 score of 0.898 on an entirely unseen dataset sourced from raw home blogs—without single-shot prompt tweaking or fine-tuning—validates the radical robustness of a neuro-symbolic approach applied to procedural knowledge.

Why EaSE-TSD Cannot Be Used for Table Setting Generalisation

What is EaSE-TSD?

The **Everyday Activity Science and Engineering Table Setting Dataset** (Meier et al., 2026, *Scientific Data* 13:721) is a multimodal biosignal corpus recorded from **78 human participants** setting a table in a controlled lab at the University of Bremen (EASE CRC).

Each session captures **eight synchronised sensor streams**:

- Marker-based **motion capture** (OptiTrack suit)
- **EEG**, **EMG**, **EDA** (electrodermal activity)
- **Eye-tracking + accelerometer**
- **First-person + environment video cameras**
- **Think-aloud microphones**

Annotated at three tiers of granularity: **phases** → **activities** → **motions**, with interacted objects tracked via knowledge-supported video analysis.

Why it cannot be used?

1. **Wrong data modality**
   EaSE-TSD contains **time-series biosignals and motion trajectories**, not discrete tableware selection labels. There is no structured column of the form `(recipe, plate_type, cutlery_set)` that could serve as ground truth for the pipeline's output.
2. **Wrong task framing**
   The dataset studies **how humans execute** a table-setting task (motion planning, cognitive load, EEG correlates). The pipeline predicts **which items to place** given a meal name. These are orthogonal research questions.

Conclusions

Complete Results: In-Distribution vs. OOD

In-Distribution — RoboCSK Benchmark

Task	Format	Full Pipeline	Pure LLM	Pure Logic	Winner
Tidy Up	MC (383 Q)	**69.97%**	60.05%	31.33%	Full +9.9 pp
Tool Usage	MC (291 Q)	**81.79%**	58.08%	28.52%	Full +23.7 pp
Table Setting	Plate (100 rec)	**88.0%**	79.0%	—	Full +9.0 pp
Table Setting	Cutlery Jacc.	**0.778**	0.691	—	Full +8.7 pp
Proc. Knowledge	Binary (3104 Q)	0.950 F1	**0.983 F1**	—	Pure LLM +0.033
Meta-Reasoning	Binary (in-dist)	0.809 F1	0.517 F1	**0.996 F1**	Pure Logic

Out-of-Distribution — Generalization Study

Task	OOD Dataset	Best (Full Pipeline)	Best (Pure LLM)	Δ Full vs. LLM
Tidy Up	VirtualHome (67 Q)	91.04%	**92.54%**	**−1.5 pp**
Tidy Up	TidyBot (598 Q)	**47.99%**	39.13%	**+8.9 pp**
Tool Usage	PIQA filtered (710 Q)	91.69%	**94.37%**	**−2.7 pp**
Meta-Reas.	BEHAVIOR-1K (2032 Q)	**0.768 F1**	0.571 F1	**+0.197 F1**
Proc. Know.	RecipeNLG (2000 Q)	**0.898 F1**	0.786 F1	**+0.112 F1**

Finding 1: The Neuro-Symbolic Pipeline Excels at Entity Selection

The full pipeline has two distinct symbolic components:

- **Knowledge Graph (KG) lookup** — ConceptNet + CSKG graph extraction: used for **Tidy Up** and **Tool Usage**. Retrieves object nodes (locations, affordances) to ground the LLM's answer space.
- **Prolog reasoner with manually engineered rules** — deterministic rule-based inference, no KG lookup: used for **Table Setting**, **Meta-Reasoning**, and **Procedural Knowledge**.

The symbolic gain is strongest when each component aligns with the task's answer structure:

Strongest gains on RoboCSK:

Task	Gain	Symbolic component
Tool Usage	**+23.7 pp**	Knowledge graph affordances filter implausible distractors
Table Setting plate	**+9.0 pp**	Prolog rules encode meal-type $\rightarrow$ plate mappings directly
Tidy Up MC	**+9.9 pp**	Knowledge graph `AtLocation` triples anchor spatial reasoning

Finding 1: The Neuro-Symbolic Pipeline Excels at Entity Selection

Size equalization effect: mistral-small + Full Pipeline (**81.79%**) outperforms mistral-large Pure LLM (**58.08%**) on Tool Usage — KG grounding compensates for model capacity when the symbolic signal is strong.

Both components generalise zero-shot under domain shift:

OOD Dataset	Δ Full vs. Pure LLM	Symbolic component
BEHAVIOR-1K	**+19.7 pp F1**	Prolog capability rules
TidyBot	**+8.9 pp**	Knowledge graph `AtLocation` triples
RecipeNLG	**+11.2 pp F1**	Prolog phase-ordering rules

All three generalise with **zero pipeline modifications**: KG triples encode domain-general location facts; Prolog rules encode structural constraints (capability axioms, phase orderings) that are task-geometry-invariant.

Finding 1: The Neuro-Symbolic Pipeline Excels at Entity Selection

Key insight: generalization comes from two independent sources — **Knowledge Graph triples** (location/affordance facts that hold across household domains) and **Prolog rules** (structural constraints invariant to writing style or activity category). BEHAVIOR-1K introduces 1,016 entirely new task categories and the full pipeline still gains +19.7 pp F1 over the pure LLM, with zero code changes.

Finding 2: The Neuro-Symbolic Pipeline Degrades on Relational & Procedural Tasks

Three distinct failure modes identified:

Failure Mode 1 — Task geometry mismatch (PIQA)

The pipeline was designed for entity selection: it retrieves KB nodes by name and injects them as exemplars. PIQA asks which **procedure** is physically correct, the correct answer is free text, never a named object. The KB mention-differential **penalises** creative correct answers and **rewards** wrong tool-like answers that happen to appear in the KB.

Result: -2.7 pp (filtered) to -6.0 pp (unfiltered) versus pure LLM. Pure Logic at 49.3% $\approx$ random chance.

Finding 2: The Neuro-Symbolic Pipeline Degrades on Relational & Procedural Tasks

Failure Mode 2 — LLM ceiling effect (VirtualHome, vocab-aligning mode)

VirtualHome uses compound or camelCase object names (`kitchencabinet`, `AlarmClock`) that do not match Knowledge Graph nodes. A **vocab-aligning preprocessing step** was added: names are split and lowercased (`kitchencabinet` → `kitchen cabinet`) to bridge VH vocabulary to KG-compatible tokens before the Prolog lookup. Even with this adaptation the LLM is already near ceiling (92%): common-sense locations (beer→fridge, alarm clock→nightstand) are high-frequency associations memorised in pre-training. The KG lookup adds no new information and $\Delta = -1.5$ pp.

Failure Mode 3 — Vocabulary bottleneck (VH strict mode)

In strict mode **no vocab normalization** is applied: raw VH names are passed directly to the KG lookup. `has_location(kitchencabinet, ...)` finds no matching KG node → the reasoner returns an empty fact set and is effectively blind. The LLM must act alone, but the pipeline prompt structure differs from Pure LLM, causing a −16.4 pp drop for mistral-small. This is a recoverable engineering issue (extend the normalisation dictionary), not a fundamental architectural failure.

Finding 2: The Neuro-Symbolic Pipeline Degrades on Relational & Procedural Tasks

Failure Mode 4 — Prolog reasoner redundant in-distribution

For Procedural Knowledge and Meta-Reasoning the **Prolog reasoner** is the active symbolic component. In-distribution it is a liability — yet the OOD results show a full reversal:

Task	In-dist. best	OOD best	Why the inversion?
Proc. Know. binary	Pure LLM 0.983 F1	Full Pipeline +0.112 F1	In-dist.: LLM has strong culinary priors; Prolog rules add rigidity. OOD (RecipeNLG): colloquial writing degrades LLM confidence; Prolog phase-ordering rules become the structural anchor
Meta-Reas. in-dist.	Pure Logic 0.996 F1	Full Pipeline +0.197 F1	In-dist.: Prolog axioms alone fully encode capability rules; LLM adds hallucination variance. OOD (BEHAVIOR-1K): novel activities require LLM understanding;Prolog anchors capability reasoning

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